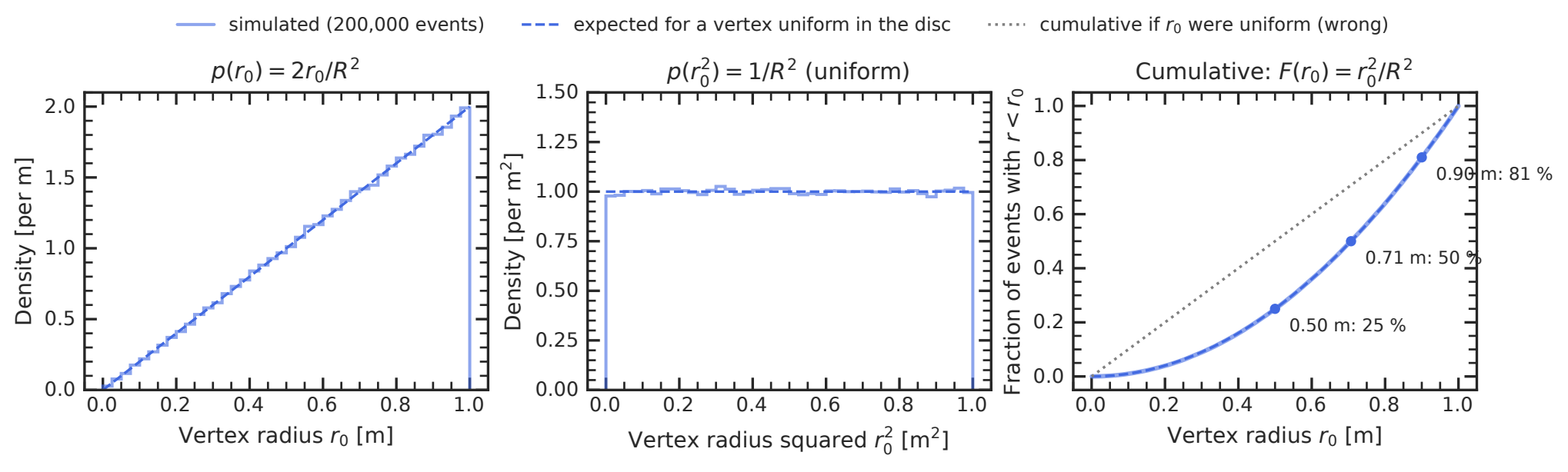
simulated (200,000 events)
expected for a vertex uniform in the disc
cumulative if r_0 were uniform (wrong)
$p(r_0) = 2r_0/R^2$
Density [per m]
Vertex radius r_0 [m]
$p(r_0^2) = 1/R^2$ (uniform)
Density [per m^2]
Vertex radius squared r_0^2 [m^2]
Cumulative: $F(r_0) = r_0^2/R^2$
Fraction of events with $r < r_0$
Vertex radius r_0 [m]
0.50 m: 25 %
0.71 m: 50 %
0.90 m: 81 %